

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2025
MA2002-1 – NUMERICAL METHODS
 (For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is not correct _____
 A) $E^{-1} = 1 - \nabla$ B) $E^{-1} + \nabla = 1$
 C) $E = 1 + \nabla$ D) $E - 1 = \Delta$
- For the given distributed data, the value of $\Delta^3 y_0 =$ _____

x	0	1	2	3
y	1	4	6	12

 A) 5 B) 3
 C) -1 D) 4
- The n^{th} order forward difference of n^{th} degree polynomial is _____
 A) a constant B) zero
 C) a polynomial of degree n D) a polynomial of degree $n - 1$
- While using Newton's forward difference interpolation formula, for any x the value of p is _____
 A) $\frac{x+x_0}{h}$ B) $\frac{x+x_n}{h}$
 C) $\frac{x-x_0}{h}$ D) $\frac{x-x_n}{h}$
- If $x_0, x_1, x_2, \dots, x_n$ are equally spaced values of x , then expression for $\frac{dy}{dx}$ at any point $x = x_n + ph$ using backward difference operator is _____
 A) $\frac{1}{h} \{ \nabla y_n + \frac{(2p-1)}{2} \nabla^2 y_n + \frac{(3p^2-6p+2)}{6} \nabla^3 y_n + \dots \}$ B) $\frac{1}{h} \{ \nabla y_n + \frac{(2p+1)}{2} \nabla^2 y_n + \frac{(3p^2+6p+2)}{6} \nabla^3 y_n + \dots \}$
 C) $\frac{1}{h} \{ \nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n + \dots \}$ D) $\frac{1}{h} \{ \nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots \}$
- Given

x	0	1	2
y	2	4	8

 the value of $y'(2) =$ _____
 A) 3 B) 6
 C) 2 D) 5
- The value of $\int_0^2 \frac{1}{1+x} dx$ _____ calculated by using Trapezoidal rule with $h=1$
 A) 1.0986 B) 1.0666
 C) 1.1666 D) 0.9986
- To apply Simpson's $3/8^{\text{th}}$ rule to get an approximate value of $\int_{x_0}^{x_n} f(x) dx$, the number of sub intervals must be a
 A) multiple of 2 B) multiple of 4
 C) multiple of 3 D) multiple of 5

9. Using Modified Euler's method, for the initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$, the first modified value of y_1 at $x = x_1$ is _____
- A) $y_1^{(1)} = y_0 + \frac{1}{2}[f(x_0, y_0) + f(x_1, y_1^{(0)})]$ B) $y_1^{(1)} = y_0 + [f(x_0, y_0) + f(x_1, y_1^{(0)})]$
 C) $y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(0)})]$ D) $y_1^{(1)} = y_0 + h[f(x_0, y_0) + f(x_1, y_1^{(0)})]$
10. In Runge-Kutta method of order four, the value of k_1 for the initial value problem $\frac{dy}{dx} = x^2 + y$, $y(0) = 1$, $h = 0.1$ is _____
- A) 0.1 B) 0.05
 C) 0.2 D) 0.25
11. Using Euler's method for the initial value problem $\frac{dy}{dx} = x - y^2$, $y(0) = 1$, $h = 0.1$ the value of y_1 is _____
- A) 0.1 B) 1.1
 C) 1.05 D) 0.9
12. Milne's predictor formula to find y_4 is _____
- A) $y_4 = y_0 + \frac{4h}{3}(f_1 - 2f_2 + f_3)$ B) $y_4 = y_1 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$
 C) $y_4 = y_0 + \frac{h}{3}(2f_1 - f_2 + 2f_3)$ D) $y_4 = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$
13. The Laplace's equation $u_{xx} + u_{yy} = 0$ is classified as _____
- A) Elliptic B) Parabolic
 C) Cyclic D) Hyperbolic
14. While finding the solution of heat equation $u_t = c^2 u_{xx}$, if the mesh ratio parameter $\alpha = \frac{1}{2}$ then $u_{i,j+1}$ is _____
- A) $\frac{1}{3}[u_{i-1,j} + u_{i+1,j}]$ B) $\frac{1}{2}[u_{i-1,j} + u_{i+1,j}]$
 C) $\frac{1}{2}[u_{i-1,j} - u_{i+1,j}]$ D) $\frac{1}{4}[u_{i-1,j} + u_{i+1,j}]$
15. For the heat equation $u_t = u_{xx}$, $h = \frac{1}{3}$, $k = \frac{1}{36}$, the value of mesh ratio parameter α is _____
- A) $\frac{1}{2}$ B) $\frac{1}{3}$
 C) $\frac{1}{4}$ D) $\frac{1}{36}$
16. For the wave equation $u_{tt} = u_{xx}$, $h = \frac{1}{4}$, $k = \frac{1}{5}$, the value of α is _____
- A) 0.5 B) 0.8
 C) 0.25 D) 1.25
17. Root of the equation $x^3 - 4x + 9 = 0$ lies in the interval,
- A) (0,1) B) (-1,-2)
 C) (-2,3) D) (-2, -3)
18. The method which approximates curve as chord to find root of an algebraic equation is _____
- A) Runge-Kutta method B) Modified Euler's method
 C) Newton-Raphson method D) Regula-Falsi method
19. In Newton-Raphson method the formula to find the 1st approximation to the root of $f(x) = 0$ is $x_1 =$ _____
- A) $x_0 - \frac{f(x_0)}{f'(x_0)}$ B) $x_1 + \frac{f(x_1)}{f'(x_1)}$
 C) $x_1 - \frac{f'(x_1)}{f(x_1)}$ D) $x_0 - \frac{f'(x_0)}{f''(x_0)}$
20. The first approximation to the root of $x^3 - 2x - 5 = 0$ in (2, 3) using bisection method is _____
- A) 2.75 B) 2.5
 C) 2.35 D) 2.25

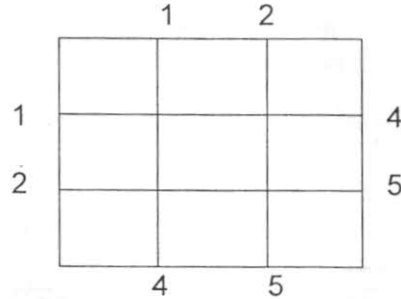
PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

1. a) Estimate the number of students who obtained marks between 40 and 45, given
- | | | | | | |
|-----------------|-------|-------|-------|-------|-------|
| Marks | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
| No. of students | 31 | 42 | 51 | 35 | 31 |
- b) Evaluate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.2$, given
- | | | | | |
|-----|------|------|------|------|
| x | 1.0 | 1.2 | 1.4 | 1.6 |
| y | 2.72 | 3.32 | 4.06 | 4.96 |
- c) Evaluate $\int_1^{2.1} \frac{1}{x} dx$, using Trapezoidal rule by considering five ordinates.
2. a) The velocity $v(\text{km/min})$ of a moped which starts from rest is given at fixed intervals of time $t(\text{min})$ as follows
- | | | | | | | | | | | |
|-----|----|----|----|----|----|----|----|----|----|----|
| t | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 |
| v | 10 | 18 | 25 | 29 | 32 | 20 | 11 | 5 | 2 | 0 |
- Using Simpson's $1/3^{\text{rd}}$ rule estimate the distance covered in 20min.
- b) Using suitable interpolation formula, find the value of x when $y = 0$ from the given data
- | | | | | |
|-----|-----|-----|----|----|
| x | 30 | 34 | 38 | 42 |
| y | -30 | -13 | 3 | 18 |
- c) Solve for $\frac{dy}{dx}$ at $x = 0$, given
- | | | | | | | |
|-----|---|---|----|---|---|---|
| x | 0 | 1 | 2 | 3 | 4 | 5 |
| y | 4 | 8 | 15 | 7 | 6 | 2 |
3. a) Apply Newton's divided difference formula, to find the value of y when $x = 1$ from the given data
- | | | | | | |
|-----|-----|-----|-----|-----|------|
| x | -3 | -1 | 0 | 3 | 5 |
| y | -30 | -22 | -12 | 330 | 3458 |
- b) Derive the Newton's forward difference interpolation formula.
- c) Evaluate $\int_0^1 \frac{dx}{1+x^2}$, using Weddle's rule by considering seven ordinates.

Unit – II

4. a) Use Adams-Bashforth method to find y at $x = 0.4$, given
- $$\frac{dy}{dx} = \frac{xy}{2}, \quad y(0) = 1, \quad y(0.1) = 1.0025, \quad y(0.2) = 1.0101, \quad y(0.3) = 1.0228.$$
- b) Solve the boundary value problem $u_t = 4u_{xx}$ under the conditions $u(x, 0) = 4x - \frac{x^2}{2}$, $u(0, t) = u(8, t) = 0$ with $h = 1$, $k = \frac{1}{8}$. Find the values upto $t = \frac{1}{2}$.
- c) Using Modified Euler's method, find an approximate value of y when $x = 0.1$, taking $h = 0.1$ for $\frac{dy}{dx} = x^2 + y$, $y(0) = 1$. Carry out two iterations in each step.

5. a) Solve $\nabla^2 u = -81xy$, $0 < x, y < 1$, $h = k = 1/3$,
 $u(x, 0) = u(0, y) = 0$, $u(x, 1) = 100 = u(1, y)$. 6 L2 4 1
- b) Use Milne's predictor-corrector method to find y at $x = 0.4$ given
 $\frac{dy}{dx} = 1 + xy^2$, $y(0) = 1$, $y(0.1) = 1.105$, $y(0.2) = 1.223$,
 $y(0.3) = 1.355$. 5 L2 3 1
- c) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown below:



6. a) Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ with the boundary conditions $u(0, t) = u(1, t) = 0$
and $u(x, 0) = 100 \sin \pi x$ and $u_t(x, 0) = 0$. Take $h = k = \frac{1}{4}$. Find the values up to $t=1$. 6 L2 4 1
- b) Apply fourth order Runge-Kutta method to find the value of y at
 $x = 0.2$, given that $\frac{dy}{dx} = 1 + y^2$, $y(0) = 0$ and $h = 0.2$. 5 L3 3 1
- c) Using Taylor's series method of order four, compute y at $x = 0.1$
Given $\frac{dy}{dx} = xy + 1$, $y(0) = 1$, take $h = 0.1$. 5 L2 3 1

Unit – III

7. a) Using Gauss Seidel method, obtain the solution of the linear system of equations taking the initial approximation to the solution as $x^{(0)} = 0, y^{(0)} = 0, z^{(0)} = 0$. Carry out three iterations.
 $83x + 11y - 4z = 95$
 $7x + 52y + 13z = 104$
 $3x + 8y + 29z = 71$ 6 L3 5 1
- b) Using Newton Raphson's method, obtain the root of the equation $x^4 - x - 10 = 0$ near $x = 2$. Carry out three iterations. 6 L2 5 1
- c) Using Ordinary iteration method obtain the root of $\cos x = 3x - 1$ near $x = 0$. Carry out four iterations. 4 L2 5 1
8. a) Using Newton Raphson method solve the following system of nonlinear equations
 $x^2 + y = 11$, $x + y^2 = 7$ with $x_0 = 3.5, y_0 = -1.8$. Find x_1 and y_1 . 6 L3 5 1
- b) Using the Regula-Falsi method find a root of the equation $xe^x = \cos x$ in $(0, 1)$. Carry out three iterations. 6 L2 5 1
- c) Using bisection method obtain the root of $x^3 + 5x - 4 = 0$ in $(0, 1)$. Carry out 4 iterations. 4 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
